1 Gain law

($e_\theta = \theta - \hat{\theta}$; coefficients at the estimate)

$$\bar{a}_{\text{true}}(\bar{w}) = \frac{1}{c_\perp(\bar{w})}$$
$$u(\bar{w}) := \frac{1}{1 - \bar{a}_{\text{true}}(\bar{w})}$$
$$b_{\text{true}}(\bar{w}) := \frac{\mathrm{d}u}{\mathrm{d}\bar{w}} = \frac{\bar{a}'_{\text{true}}(\bar{w})}{\big(1 - \bar{a}_{\text{true}}(\bar{w})\big)^2}$$
$$\bar{a}'_{\text{true}}(\bar{w}) = b_{\text{true}}(\bar{w})\big(1 - \bar{a}_{\text{true}}(\bar{w})\big)^2$$
$$\lim_{\bar{w}\to\infty} b_{\text{true}} = 1, \qquad \lim_{\bar{w}\to 1} b_{\text{true}} = \frac{8}{9}$$

2 Model

$$b_{\text{true}}(\bar{w}) \longrightarrow \hat{b} \quad \text{(constant)}$$
$$\Delta b(\bar{w}) := b_{\text{true}}(\bar{w}) - \hat{b}$$

3 Step

$$\frac{1}{1 - \hat{\bar{a}}_w[k+1]} = \frac{1}{1 - \hat{\bar{a}}_w[k]} + \hat{b}[k]\,\Delta\hat{\bar{w}}[k]$$
$$\frac{1}{1 - \bar{a}_{\text{true}}[k+1]} = \frac{1}{1 - \bar{a}_{\text{true}}[k]} + \int_{\bar{w}[k]}^{\bar{w}[k]+\Delta\bar{w}[k]} b_{\text{true}}(\bar{w})\,\mathrm{d}\bar{w}$$

4 Law error of one step

$$\frac{\partial\,\hat{\bar{a}}_w[k+1]}{\partial\,\hat{b}[k]} = \big(1 - \hat{\bar{a}}_w[k+1]\big)^2\,\Delta\hat{\bar{w}}[k] = \Big(\frac{1 - \hat{\bar{a}}_w[k+1]}{1 - \hat{\bar{a}}_w[k]}\Big)^2\big(1 - \hat{\bar{a}}_w[k]\big)^2\,\Delta\hat{\bar{w}}[k]$$
$$\varepsilon_{\text{law}}[k] = \big(1 - \hat{\bar{a}}_w[k]\big)^2\,\Delta b\big(\bar{w}[k]\big)\,\Delta\hat{\bar{w}}[k] + \mathcal{O}\big(\Delta\hat{\bar{w}}^2\big)$$

5 What the covariance already carries

$$F_e[k]_{4,5} = \big(1 - \hat{\bar{a}}_w[k]\big)^2\big(\Delta\bar{w}_d[k] + (1 - \lambda_c)\,\delta\hat{\bar{w}}_3[k]\big)$$
$$\big(F_e[k]\,\mathbf{e}[k]\big)_4 \supset \big(1 - \hat{\bar{a}}_w[k]\big)^2\,\Delta\hat{\bar{w}}[k]\,e_b[k]$$

$$b_{\text{true}}(\bar{w}) = \hat{b} + e_b + \Delta b(\bar{w}) - e_b$$

carried: e_b (which constant)

not carried: $\Delta b(\bar{w}) - e_b$ (no constant fits the curve)

6 Envelope

$$\delta_b := \sup_{\bar{w}\in\text{env}} \left| b_{\text{true}}(\bar{w}) - \hat{b} \right|$$

$$b_{\text{true}}(\bar{w}) \in \left[\tfrac{8}{9},\ 1\right] \quad\Longrightarrow\quad \delta_b \le \tfrac{1}{9}$$

$$\sqrt{P_{55}[0]} = \sup_{\bar{w}\in\text{env}} \left| b_{\text{true}}(\bar{w}) - \tfrac{8}{9} \right|$$

7 Injection

$$\Delta Q_{44}[k] = \Big(\big(1 - \hat{\bar{a}}_w[k]\big)^2 \, \delta_b \, \Delta\hat{\bar{w}}[k] \Big)^2$$

$$\Delta Q_{45}[k] = \Delta Q_{54}[k] = 0$$

$$Q[k] \leftarrow Q[k] + \Delta Q_{44}[k]\, \mathbf{e}_4 \mathbf{e}_4^\top$$

$$\Delta\hat{\bar{w}}[k] = 0 \quad\Longrightarrow\quad \Delta Q_{44}[k] = 0$$

8 Colour

$$\sum_{k=k_0}^{k_0+N-1} \varepsilon_{\text{law}}[k] = \big(1 - \hat{\bar{a}}_w\big)^2 \int_{\bar{w}[k_0]}^{\bar{w}[k_0+N]} \Delta b(\bar{w})\, \mathrm{d}\bar{w}$$

Δb of one sign over the path $\quad\Longrightarrow\quad \left|\sum \varepsilon_{\text{law}}\right| = N \left|\varepsilon_{\text{law}}\right|$

white injection $\quad\Longrightarrow\quad \sqrt{\mathbb{E}\Big[\big(\sum \varepsilon_{\text{law}}\big)^2\Big]} = \sqrt{N}\left|\varepsilon_{\text{law}}\right|$

$$\frac{\text{deterministic}}{\text{white}} = \sqrt{N}$$